Given ISA Round Robin (Q=3)

Process	Arrival time (AT)	Burst time (BT)
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5
P_5	5	2

time Quantum = 3

P_1 0-3
P_2 3-6
P_3 6-9
P_4 9-12
P_1 12-15
P_5 15-17
P_2 17-18
P_3 18-21
P_4 21-23
P_1 23-25
P_3 25-28

Completion time (CT)

$CT(P_1) = 25$
$CT(P_2) = 18$
$CT(P_3) = 28$
$CT(P_4) = 23$
$CT(P_5) = 17$

$TAT = CT - AT$
$TAT(P_1) = 25 - 0 = 25$
$TAT(P_2) = 18 - 1 = 17$
$TAT(P_3) = 28 - 2 = 26$
$TAT(P_4) = 23 - 3 = 20$
$TAT(P_5) = 17 - 5 = 12$

Round Robin (Q=3)

SA

Wating time (WT)

WT = TAT - BT

WT(P1) = 25-8 = 17
WT(P2) = 17-4 = 13
WT(P3) = 26-9 = 17
WT(P4) = 20-5 = 15
WT(P5) = 12-2 = 10

Response time (RT)

RT = first start time - AT

RT(P1) = 0-0 = 0
RT(P2) = 3-1 = 2
RT(P3) = 6-2 = 4
RT(P4) = 9-3 = 6
RT(P5) = 15-5 = 10

average TAT

$$\frac{25+17+26+20+12}{5} = 20$$

average WT $= \dfrac{17+13+17+15+10}{5} = 14.4$

~~atto~~ average RT $= \dfrac{0+2+4+6+10}{5} = 4.4$

SA

2) SJF - Preemptive (SRTF)

Gantt Chart

P_1	0-1
P_2	1-5
P_5	5-7
P_4	7-12
P_1	12-19
P_3	19-28

Completion time (ct)

$ct(P_1) = 19$
$ct(P_2) = 5$
$CT(P_3) = 28$
$CT(P_4) = 12$
$CT(P_5) = 7$

Turnaround time (TAT)

$TAT = CT - AT$
$TAT(P_1) = 19-0 = 19$
$TAT(P_2) = 5-1 = 4$
$TAT(P_3) = 28-2 = 26$
$TAT(P_4) = 12-3 = 9$
$TAT(P_5) = 7-5 = 2$

waiting time (wT)

$wT = TAT - BT$
$wT(P_1) = 19-8 = 11$
$wT(P_2) = 4-4 = 0$
$wT(P_3) = 26-9 = 17$
$wT(P_4) = 9-5 = 4$
$wT(P_5) = 2-2 = 0$

Response time (RT)

$RT = \text{First time} - AT$

$RT(P_1) = 0-0 = 0$
$RT(P_2) = 1-1 = 0$
$RT(P_3) = 19-2 = 17$
$RT(P_4) = 7-3 = 4$
$RT(P_5) = 5-5 = 0$

Average TAT

$$= \frac{19+4+26+\cancel{8}9+2}{5} = 12$$

Average wT =

$$\frac{11+0+17+4+0}{5} = 6.4$$

Average RT

$$= \frac{0+0+17+4+0}{5}$$

$= 4.2$

SC

Round Robin (Q = 4)

Given

Process	Arrival time (AT)	Brust Time (BT)
P_1	0	10
P_2	0	10
P_3	0	10
P_4	0	10
P_5	0	10

Time Quantum = 4

Gantt Chart

P_1 0-4
P_2 4-8
P_3 8-12
P_4 12-16
P_5 16-20
P_1 20-24
P_2 24-28
P_3 28-32
P_4 32-36
P_5 36-40
P_1 40-42
P_2 42-44
P_3 44-46
P_4 46-48
P_5 48-50

Completion time (Ct)

$Ct(P_1) = 42$
$CT(P_2) = 44$
$Ct(P_3) = 46$
$Ct(P_4) = 48$
$Ct(P_5) = 50$

SC

Turnaround time (TAT)

TAT = CT − AT

TAT(P1) = 42 − 0 = 42

TAT(P2) = 44 − 0 = 44

TAT(P3) = 46 − 0 = 46

TAT(P4) = 48 − 0 = 48

TAT(P5) = 50 − 0 = 50

waiting time (WT)

WT = TAT − BT

WT(P1) = 42 − 10 = 32

WT(P2) = 44 − 10 = 34

WT(P3) = 46 − 10 = 36

WT(P4) = 48 − 10 = 38

WT(P5) = 50 − 10 = 40

Response time

RT = first start time − AT

RT(P1) = 0

RT(P2) = 4

RT(P3) = 8

RT(P4) = 12

RT(P5) = 16

average TAT

$$\frac{42+44+46+48+50}{5} = 46$$

Average WT $= \dfrac{32+34+36+38+40}{5} = 36$

Average RT $= \dfrac{0+4+8+12+16}{5} = 8$

SC

2) SJF - Preemptive (SRTF)

Gantt Chart

P_1 0-10

P_2 10-20

P_3 20-30

P_4 30-40

P_5 40-50

Completion time (CT)

$CT(P_1) = 10$

$CT(P_2) = 20$

$CT(P_3) = 30$

$CT(P_4) = 40$

$CT(P_5) = 50$

Turnaround time (TAT)

$TAT = CT - AT$

$TAT(P_1) = 10 - 0 = 10$

$TAT(P_2) = 20 - 0 = 20$

$TAT(P_3) = 30 - 0 = 30$

$TAT(P_4) = 40 - 0 = 40$

$TAT(P_5) = 50 - 0 = 50$

Waiting time (WT)

$WT = TAT - BT$

$WT(P_1) = 10 - 10 = 0$

$WT(P_2) = 20 - 10 = 10$

$WT(P_3) = 30 - 10 = 20$

$WT(P_4) = 40 - 10 = 30$

$WT(P_5) = 50 - 10 = 40$

Response time (RT)

$RT(P_1) = 0$

$RT(P_2) = 10$

$RT(P_3) = 20$

$RT(P_4) = 30$

$RT(P_5) = 40$

average TAT

$$\frac{10+20+30+40+50}{5} = 30$$

average WT

$$\frac{0+10+20+30+40}{5} = 20$$

average RT $= \dfrac{0+10+20+30+40}{5} = 20$

SD

Round Robin
(Q = 3)

Time Quantum = 3

Given

Process	(AT) Arrival time	(BT) Burst Time
P_1	0	20
P_2	1	3
P_3	2	2
P_4	4	4
P_5	5	1

Gantt Chart

P_1	0-3
P_2	3-6
P_3	6-8
P_1	8-11
P_4	11-14
P_5	14-15
P_1	15-18
P_4	18-19
P_1	19-22
P_1	22-25
P_1	25-28
P_1	28-30

Completion time (CT)

CT (P_1) = 30

CT (P_2) = 6

CT (P_3) = 8

CT (P_4) = 19

CT (P_5) = 15

Turnaround Time (TAT)   TAT = CT − AT

TAT (P_1) = 30 − 0 = 30

TAT (P_2) = 6 − 1 = 5

TAT (P_3) = 8 − 2 = 6

TAT (P_4) = 19 − 4 = 15

TAT (P_5) = 15 − 5 = 10

SD Round Robin

Waiting time (WT)

WT = TAT - BT

WT(P1) = 30 - 20 = 10

WT(P2) = 5 - 3 = 2

WT(P3) = 6 - 2 = 4

WT(P4) = 15 - 4 = 11

WT(P5) = 10 - 1 = 9

Response time (RT)

RT = first start time - AT

RT(P1) = 0 - 0 = 0

RT(P2) = 3 - 1 = 2

RT(P3) = 6 - 2 = 4

RT(P4) = 11 - 4 = 7

RT(P5) = 14 - 5 = 9

Average TAT

$$= \frac{30+5+6+15+10}{5} = 13.2$$

$$\text{Average WT} = \frac{10+2+4+11+9}{5} = 7.2$$

$$\text{Average RT} = \frac{0+2+4+7+9}{5} = 4.4$$

2) SJF - Preemptive (SRTF)

Gantt Chart

P_1	0-1
P_2	1-4
P_3	4-6
P_5	6-7
P_4	7-11
P_1	11-30

Completion time (CT)

$CT(P_1) = 30$
$CT(P_2) = 4$
$CT(P_3) = 6$
$CT(P_4) = 11$
$CT(P_5) = 7$

Turnaround time (TAT)

$TAT = CT - AT$
$TAT(P_1) = 30-0 = 30$
$TAT(P_2) = 4-1 = 3$
$TAT(P_3) = 6-2 = 4$
$TAT(P_4) = 11-4 = 7$
$TAT(P_5) = 7-5 = 2$

waiting time (WT)

$WT = TAT - BT$
$WT(P_1) = 30-20 = 10$
$WT(P_2) = 3-3 = 0$
$WT(P_3) = 4-2 = 2$
$WT(P_4) = 7-4 = 3$
$WT(P_5) = 2-1 = 1$

Response time (RT)

$RT = \text{first start} - AT$
$RT(P_1) = 0-0 = 0$
$RT(P_2) = 1-1 = 0$
$RT(P_3) = 4-2 = 2$
$RT(P_4) = 7-4 = 3$
$RT(P_5) = 6-5 = 1$

$$\text{average TAT} = \frac{30+3+4+7+2}{5} = 9.2$$

$$\text{average WT} = \frac{10+0+2+3+1}{5} = 3.2$$

$$\text{average RT} = \frac{0+0+2+3+1}{5} = 1.2$$

RR

Quantum = 2

Process	Arrival Time (AT)	Burst Time (BT)
P_1	0	1
P_2	0	2
P_3	0	1
P_4	0	3
P_5	0	1

1 Gantt chart

P1: 0 → 1
P2: 1 → 3
P3: 3 → 4
P4: 4 → 6
P5: 6 → 7
P4: 7 → 8

2 Completion Time (CT)

CT(P1) = 1
CT(P2) = 3
CT(P3) = 4
CT(P4) = 8
CT(P5) = 7

3 Turnaround time (TAT) CT − AT

TAT(P1) = 1 − 0 = 1
TAT(P2) = 3 − 0 = 3
TAT(P3) = 4 − 0 = 4
TAT(P4) = 8 − 0 = 8
TAT(P5) = 7 − 0 = 7

$$\boxed{\text{Average TAT} = \frac{(1+3+4+8+7)}{5} = 4.6}$$

4 Waiting Time (WT) TAT − BT

WT(P1) = 1 − 1 = 0
WT(P2) = 3 − 2 = 1
WT(P3) = 4 − 1 = 3
WT(P4) = 8 − 3 = 5
WT(P5) = 7 − 1 = 6

$$\boxed{\text{Average WT} = \frac{(0+1+3+5+6)}{5} = 3}$$

5 Response Time (RT) = first start Time − AT

RT(P1) = 0 − 0 = 0
RT(P2) = 1 − 0 = 1
RT(P3) = 3 − 0 = 3
RT(P4) = 4 − 0 = 4
RT(P5) = 6 − 0 = 6

$$\boxed{\text{Average RT} = \frac{(0+1+3+4+6)}{5} = 2.8}$$

2 (SRTF)

1 Gantt Chart

P1: 0 → 1

P2: 1 → 2

P5: 2 → 3

P2: 3 → 5

P4: 5 → 8

2 Completion Time (CT)

CT (P1) = 1

CT (P2) = 5

CT (P3) = 2

CT (P4) = 8

CT (P5) = 3

3 Turnaround Time (TAT) = CT - AT

TAT (P1) = 1 - 0 = 1

TAT (P2) = 5 - 0 = 5

TAT (P3) = 2 - 0 = 2

TAT (P4) = 8 - 0 = 8

TAT (P5) = 3 - 0 = 3

$$\boxed{\text{Average TAT} = \frac{(1+5+2+8+3)}{5} = 3.8}$$

4 Waiting time (WT) = TAT - BT

WT (P1) = 1 - 1 = 0

WT (P2) = 5 - 2 = 3

WT (P3) = 2 - 1 = 1

WT (P4) = 8 - 3 = 5

WT (P5) = 3 - 1 = 2

$$\boxed{\text{Average WT} = \frac{(0+3+1+5+2)}{5} = 2.2}$$

5 Response Time (RT) = first start time - AT

RT (P1) = 0 - 0 = 0

RT (P2) = 3 - 0 = 3

RT (P3) = 1 - 0 = 1

RT (P4) = 5 - 0 = 5

RT (P5) = 2 - 0 = 2

$$\boxed{\text{Average RT} = \frac{(0+3+1+5+2)}{5} = 2.2}$$

RR

process	Arrival Time (At)	Burst Time (Bt)
P_1	0	20
P_2	1	1
P_3	1	3
P_4	2	2
P_5	3	1
P_6	4	2

Quantum = 2

1 Gant chart

P_1 0 → 2
P_2 2 → 3
P_3 3 → 5
P_4 5 → 7
P_1 7 → 9
P_5 9 → 10
P_6 10 → 12
P_3 12 → 13
P_1 13 → 29

2 Completion time (CT)

CT(p1) = 29
CT(p2) = 3
CT(p3) = 13
CT(p4) = 7
CT(p5) = 10
CT(p6) = 12

3 Turnaround Time (TAT) CT - AT

TAT(p1) = 29 - 0 = 29
TAT(p2) = 3 - 1 = 2
TAT(p3) = 13 - 1 = 12
TAT(p4) = 7 - 2 = 5
TAT(p5) = 10 - 3 = 7
TAT(p6) = 12 - 4 = 8

$\boxed{\text{Average TAT} = (29+2+12+5+7+8)/6 = 10.5}$

4 Waiting Time (WT) = TAT - BT

WT (P1) = 29 - 20 = 9

WT (P2) = 2 - 1 = 1

WT (P3) = 12 - 3 = 9

WT (P4) = 5 - 2 = 3

WT (P5) = 7 - 1 = 6

WT (P6) = 8 - 2 = 6

$\boxed{\text{Average WT} = (9+1+9+3+6+6)/6 = 5.67}$

5 Response Time (RT) first start time - AT

RT (P1) = 0 - 0 = 0

RT (P2) = 2 - 1 = 1

RT (P3) = 3 - 1 = 2

RT (P4) = 5 - 2 = 3

RT (P5) = 9 - 3 = 6

RT (P6) = 10 - 4 = 6

$\boxed{\text{Average RT} = (0+1+2+3+6+6)/6 = 3}$

2 (SRTF)

1 Gant chart

P1 : 0 → 1
P2 : 1 → 2
P4 : 2 → 3
P5 : 3 → 4
P4 : 4 → 5
P6 : 5 → 7
P3 : 7 → 10
P1 : 10 → 29

2 Completion Time (CT)

CT (P1) = 29
CT (P2) = 2
CT (P3) = 10
CT (P4) = 5
CT (P5) = 4
CT (P6) = 7

3 Turnaround Time (TAT)

TAT (P1) = 29 - 0 = 29
TAT (P2) = 2 - 1 = 1
TAT (P3) = 10 - 1 = 9
TAT (P4) = 5 - 2 = 3
TAT (P5) = 4 - 3 = 1
TAT (P6) = 7 - 4 = 3

Average TAT = $\frac{(29+1+9+3+1+3)}{6} = 7.6$

4 waiting Time (WT) TAT - BT

WT (P1) = 29 - 20 = 9
WT (P2) = 1 - 1 = 0
WT (P3) = 9 - 3 = 6
WT (P4) = 3 - 2 = 1
WT (P5) 1 - 1 = 0
WT (P6) 3 - 2 = 1

Average WT = $\frac{9+0+6+1+0+1}{6} = 2.8$

5 Response Time First start Time - AT

RT (P1) = 0 - 0 = 0
RT (P2) = 1 - 1 = 0
RT P(3) = 7 - 1 = 6
RT (P4) = 2 - 2 = 0
RT (P5) = 3 - 3 = 0
RT (P6) = 5 - 4 = 1

Average RT = $\frac{0+0+6+0+0+1}{6} = 1.3$